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(57) Abstract :
ABSTRACT Health monitoring has become something of extreme importance for all the countries in the world due to the advent of Covid-19., During the pandemic, the hospitals were flooded with patients and nurses, health officials were unable to monitor all the patients continuously. Due to this hectic schedule, many nurses even forgot to change pr stop the drip at the right time. With the constant advancement in technology, the medical field is always looking for modern electronic devices for easier detection of irregularities within the body. To overcome these problem's, loT based health monitoring approach is a need of the hour. The Internet of Things (loT) is a fast-growing technology in which the entities involved are connected to the internet for effective communication between them. The Internet of things serves as a catalyst in the healthcare field and plays a significant role in remote healthcare monitoring. With the sudden boost in the use of wearable sensors and smartphones, these remote healthcare monitoring devices have started evolving at a steady pace. In this work, we propose an IoT-based patient health monitoring system that includes health condition prediction of the patient and smart Intravenous (IV) set management. The system starts with measuring vital information like body temperature, pulse rate, Electrocardiogram (ECG), oxygen saturation, blood pressure, and glucose content continually with the help of sensors attached to the patients. This information is transferred to the monitoring device with the help of a cloud server. By using this system, the future health condition of the patient can also be predicted using the time-series data received from the wearable sensors. In order to improve the prediction result, we have used Random Forest Model boosted by ARIMA (Autoregressive Integrated Moving Average) method. This methodology also proposes a smart IV technique that detects the saline level, generates an alert of various categories, and stops the flow when the bottle becomes empty. With the help of this smart IV system, the staff can monitor the saline status remotely and thereby avoiding the blood being sucked into the container when the saline solution is not stopped at the right time. Raspberry Pi microcontroller plays a major role in collecting the sensor data, transferring the data to the cloud server, and automating the IV drip. By comparing the results with existing commercially available devices, this product will produce more accurate results. This technique may potentially be valuable in saving lives in the pandemic situations like covid-19.

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